ALESSANDRO PUCA

PLANT PATHOLOGIST

CONTACT

+39 3488763027

alessandropuca@cnr.it

Prato, Po, 59100, Italy

www.linkedin.com/in/fanofgodzillamovies

https://orcid.org/0000-0002-4587-8912

HARD-SKILLS

- Expertise in **Molecular Biology & Diagnostics**: digital PCR, RT-qPCR, LAMP, CRISPR Cas13-based detection.
- **RNA and DNA extraction** from different matrices (wood, soil, plant material).
- **Phytopathological surveys for quarantine pathogens** (e.g., *Xylella fastidiosa*, *Phytophthora* spp., *Erwinia amylovora*, *Fusarium circinatum*).
- **Biochemistry & Microbiology** - Enzymatic assays for fungal degradation activity, microorganisms (fungi, bacteria) isolation and preservation, and physiological analyses of cultures.
- Grapevine **in vitro propagation.**
- **Statistical and computational analysis in R and RStudio.** Tidyverse proficiency (dplyr, tidyr, ggplot2, stringr)
- Image analysis with ImageJ.

LANGUAGES

- English (Fluent)
- French (Fluent)
- Italian (Mother tongue)
- German (Basic)
- Chinese (Basic)

SOFT-SKILLS

- Leadership and time management. Management of laboratory staff for research projects and students supervision.
- Public speaking for expert and non-expert audiences - Presented 4 talks and 3 posters at international congresses.
- Ability to work independently or in team.

PROFILE

Plant pathologist with a dual PhD in Agricultural and Environmental Sciences (DAGRI, University of Florence & LVBE, Université de Haute-Alsace). Expertise in molecular techniques (qPCR, digital PCR, LAMP), especially for forest invasive species detection. Experience in the field of trunk diseases and fungal wood degradation processes.

WORK EXPERIENCE

Research Fellow — 03/25 - Present

CNR – Institute for the Sustainable Protection of Plants (IPSP) - Sesto Fiorentino (FI), Italy

- Development of protocols and validation of technologies for the ultrasensitive detection of *Xylella fastidiosa* (1LIVEXylella project) through portable qPCR platforms and digital PCR.
- Side project on viability assessment of *Ceratocystis platani* through mRNA detection.
- Tasked with in-field surveys for quarantine pests for the phytosanitary service

Ph.D. in Agricultural and Environmental Sciences — 11/21 - 07/2025

University of Haute-Alsace (LVBE), Colmar - University of Florence (DAGRI)

- Advisors: Prof. Laura Mugnai, Prof. Christophe Bertsch
- Thesis title: "Enzymatic and Non-Enzymatic Wood Degradation by Basidiomycetes in the Esca Complex of Grapevine."
- Investigated the role of white rot (*Fomitiporia* spp.) wood degradation in Esca, using a multidisciplinary approach that integrated proteomics, metabolomics, FT-IR spectroscopy, microscopy, and phenotypic analysis.
- Worked in an international setting in France, Italy, Sweden (COST Action grant winner), and Germany.

Erasmus+ Internship student — 09 - 12 / 2020

Southern Swedish Forest Research Center (SLU) - Lomma, Sweden

- Study of elm endophytes as possible biocontrol agents for forest pathogens (*Ophiostoma novo ulmi*)

Internship Student — 08 - 11 / 2018

CNR - IPSP - Sesto Fiorentino (FI), Italy

- Forest pathogens detection and identification, pathogenicity trials, and physiological studies on *Ostrya carpinifolia*

EDUCATION

110/110 M.S. In Agricultural Sciences and Technologies (LM-69) — 02/2019 - 11/2021

University of Florence (DAGRI)

- Plant Medicine curriculum: specialization in sustainable plant defense (agricultural and forest systems).
- Dissertation: "Evaluation of the potential of *Monographella nivalis* var. *neglecta* as an antagonist of the pathogen *Ophiostoma novo-ulmi*."

B.S. in Forest and Environmental Science (L-25) — 9/14 - 2/2018

University of Florence (DAGRI)

- Dissertation: "Phytopathological inquiry on Hophornbeam decline in Tuscany."
- Formation of core skills such as forest pathology, botany, and ecology.

DRIVING LICENSE

- A2
- B

CONGRESSES AND TALKS

- Oral presentations as presenting author:
  - XX International Plant Protection Congress, 1-5/07/2024, Athens, Greece.
  - SIPaV XXVIII – Italian Phytopathological Society, 18-20/09/2023, Naples, Italy.
- Posters:
  - 17th International Congress of the Mediterranean Phytopathological Union, 6-10/7/2025, Bari, Italy.
  - SIPaV XXXI – Italian Phytopathological Society, 9-11/09/2024, Trento, Italy.
  - ICPP – 12th International Congress of Plant Pathology at Lyon, France, 20-25/08/2023
  - IUFRO Conference Division 7 – Forest Health Pathology and Entomology, Lisbon 6 – 9/09/2022
- Presentation at Public Outreach Events:
  - Journée d'échanges entre chercheurs et professionnels de la filière viti-vinicole, 19/04/2024, Colmar, France

ADDITIONAL COURSES AND SUMMER SCHOOLS

- **2P4S2F Plant Pathogenomics for a Sustainable Future of Food** - 23-27/2025, Bologna, Italy.
  - Week-long advanced training in bioinformatics, genome sequencing, and plant-pathogen interactions.
- **EUniWell Research Communication Workshop for Doctoral Candidates** - 18/11, 9/12/2022, 9/01/2023
  - 12-hour course on science communication to the industry sector and the public.
- **6° Plant Physiopathology School** - 5-7/12/2022, Lucca, Italy.
  - Intensive 3-day training on plant pathology and plant physiology.

OTHER WORK EXPERIENCES AND INFORMATION:

TEACHING EXPERIENCE

Temporary lecturer for Master Courses — 10 -12/2023

University of Haute-Alsace

- Molecular Aspects of Bio-Interactions (TP)
  - Five-hour course on how to perform DNA extraction, qPCRs, as well as relative data analysis;
- Plants nutrition (TP) - Six-hour course covering grapevine micropropagation on different media followed by phenotypic observations.

Intership Supervisor — 04 - 06 /2024

- M1 Master in Microbiology, Université de Lorraine
- 245-hour stage with the title: Characterization of microorganisms in the Esca complex of grapevine: focusing on the degradation capabilities of the agents responsible for the formation of white rot

Occasional Science Educator

- Freelance activity at OpenLab – Science Communication Laboratory, University of Florence..
- Conducted educational sessions in elementary, middle, and high schools, explaining and performing scientific experiments on various topics.

MAIN PUBLICATIONS

- Puca, A., Pecori, F., Pepori, A., Santini, A., Luchi, N., (*in redaction*) "**Combining Field-Deployable Molecular Diagnostic Platform and Digital PCR for Point-Of-Care (POC) Diagnostics of *Xylella fastidiosa*.**"
- Puca, A., Moretti, S., Goddard, M.-L., Lalevée, J., Kassemeyer, H.-H., Farine, S., Mugnai, L., Bertsch, C., 2025. **"Around the world in eight white rot species: Assessment of enzymatic and non-enzymatic wood decay pathways of worldwide Esca Complex of Diseases (ECD) associated basidiomycetes"**. Fungal Biol. 129, 101661.
- Moretti, S., Goddard, M.-L., Puca, A., Lalevée, J., Di Marco, S., Mugnai, L., Gelhaye, E., Goodell, B., Bertsch, C., Farine, S., 2023. **"First Description of Non-Enzymatic Radical-Generating Mechanisms Adopted by Fomitiporia mediterranea: An Unexplored Pathway of the White Rot Agent of the Esca Complex of Diseases"**. J. Fungi 9, 498.

GRANTS AND FELLOWSHIPS

- COST Action CA20132 "Urban Tree Guard" grant winner - Short-term scientific mission performed in Linnaeus University, Växjö, Sweden, during my Ph. D.
- "Young researchers in training" grant winner (XXVIII SIPaV congress)
- Winner of SIPaV financing grant for XX International Plant Protection Congress (IPPC)

VOLUNTEER WORK

Red Cross - Comitato della Piana Pistoiese — 2017 - 2023

- Volunteer work as an operator with the following certifications:
  - TS-SA, Medical transport, and ambulance rescue;
  - OPEM, CRI Emergency Activities Operator.
- Planner and responsible person for the educational vegetable garden of the "NoiCiSiamo" project.

REFERENCES

DR. NICOLA LUCHI (CURRENT EMPLOYER)
CNR - IPSP
Email : nicola.luchi@cnr.it

PROF. JOHANNA WITZELL
Department of Forestry and Wood Technology- Linneaus University
Email : johanna.witzell@lnu.se

PROF. CHRISTOPHE BERTSCH
LVBE - Université de Haute-Alsace
Email : christophe.bertsch@uha.fr